

Machine Failure Evidence Checklist

Capture useful evidence before resetting, restarting, changing configuration, or allowing the failure state to disappear.

1. Freeze the moment before reset

- ☐ Record the exact local time and timezone. Photograph or screenshot the operator display before acknowledging or clearing anything.
- ☐ Record operating mode, sequence step, machine state, recipe/configuration, active user, and current product or experiment.
- ☐ Preserve the first-out trip, initiating interlock, and every active permissive - not only the final alarm cascade.

2. Compare intent with reality

- ☐ Capture commanded values and corresponding readbacks for motion, power, valves, pumps, temperatures, pressures, flows, and timing.
- ☐ Identify values that were stale, invalid, disconnected, saturated, limited, or outside expected tolerance.
- ☐ Record controller state, PLC/IOC/application health, communications status, and relevant process exit codes.

3. Preserve sequence and timing

- ☐ Export event, alarm, historian, IOC/PLC, application, and operating-system logs for an agreed window before and after the event.
- ☐ Confirm clock synchronization and note timestamp resolution. Alarm-list order alone does not prove causality.
- ☐ Capture packet, frame, trigger, scan, cycle, retry, timeout, and dropped-sample counters where applicable.

4. Capture physical and organizational context

- ☐ Record environmental and utility conditions: temperature, humidity, cooling, vacuum, power, network, vibration, and facility state.
- ☐ Record recent maintenance, software/configuration changes, substitutions, calibrations, overrides, and unusual operator actions.
- ☐ Ask what was heard, seen, smelled, felt, or manually changed. Separate observation from interpretation.

5. Package the evidence

- ☐ Create one incident ID and collect screenshots, exports, notes, versions, and relevant files under that identifier.
- ☐ Write the expected sequence, observed sequence, first confirmed divergence, recovery action, and whether the failure repeated.
- ☐ Preserve the raw evidence. Make transformed data, filtering, exclusions, and assumptions explicit and reproducible.

Detection is not diagnosis. An alarm tells you where the machine noticed a problem. Diagnosis requires sequence, state, timing, commands, readbacks, interlocks, context, and physics.